

Fenghui Yu

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Employment

Oct 2023–Present *Assistant Professor (Tenured)*
Aug 2022–Sept 2023 *Assistant Professor (Tenure-Track)*
- Delft Institute of Applied Mathematics (DIAM), TU Delft, Netherlands

Jun 2019–July 2022 *Postdoctoral Researcher*
- RiskLab and Department of Mathematics, ETH Zürich, Switzerland

Jan 2019–May 2019 *IMR Postdoctoral Fellow*
- Department of Mathematics, The University of Hong Kong, Hong Kong

Further Positions

Sept 2026–Present *Associate Faculty*
European Laboratory for Learning and Intelligent Systems (ELLIS) Unit Delft, Netherlands

**Sept–Oct 2025
& Oct–Dec 2024** *Forschungsinstitut für Mathematik (FIM) Academic Guest*
Department of Mathematics, ETH Zürich, Switzerland

Aug 2015 *Academic Visitor*
Department of Mathematical Sciences, University of Copenhagen, Denmark

Education

Sept 2014–Dec 2018 Ph.D., Mathematics - The University of Hong Kong, Hong Kong
Thesis: On Pricing, Hedging and Trading in Financial Management

Sept 2010–Jul 2014 B.S. Mathematics; B.E. Finance (Double Degree) - Jilin University, China
Major: Computational Mathematics; Minor: Finance and Financial Management

Research Interests

- Market microstructure, including limit order books, optimal execution, and market making.
- Stochastic optimal control and its applications to financial decision-making problems.
- Reinforcement learning, including theoretical methods and applications in finance and engineering.

Fundings

2025–2027 Strategic Research Initiative of *Real-World-Inspired Sequential Decision-Making* (joint application), €20,000 - 4TU.AMI, Netherlands

2024–2025 IDEA League Fellowship (Sole PI), €21,310 - IDEA League strategic alliance
Generative AI: theoretical foundations and applications in finance

Honors and Awards

2024–2025	IDEA League Fellowship - <i>IDEA League strategic alliance</i>
2018	Scientific Research Network (WOG) Grant - <i>Research Foundation Flanders (FWO), Belgium</i>
2016, 2018	Conference Grant for Research Postgraduate Students - <i>The University of Hong Kong, Hong Kong</i>
2016	SIAM Student Travel Award - <i>SIAM, USA</i>
2016	Doris Chen Postgraduate Travel Grant - <i>The University of Hong Kong, Hong Kong</i>
2016	International Student Travel Award - <i>PIMS, Canada</i>
2014–2018	HKU Foundation Postgraduate Fellowship - <i>The University of Hong Kong, Hong Kong</i>
2014–2018	Postgraduate Scholarship - <i>The University of Hong Kong, Hong Kong</i>
2012–2014	First Rank Scholarship - <i>Jilin University, China</i>
2011–2014	Outstanding Undergraduate - <i>Jilin University, China</i>

Papers

- Yaniv Oren, Viliam Vadoz, Wiktor Zabka, Thomas Evers, Jan Robine, Wendelin Böhmmer, Matthijs T. J. Spaan, Martha White, Hendrik Baier, and Fenghui Yu. *Towards Optimal Policy Improvement*. Submitted, arxiv:2610.01566, 2026.
- Fenghui Yu. *Explicit Signal-Adaptive Sequential Optimal Execution Quotes*. Submitted, arxiv:2605.24242, 2026.
- Junhong Dai, Yue Cui, Fenghui Yu, Chao Huang, and Panshuo Li. *Reinforcement Learning-based Hierarchical Planning for Multi-Vehicle Coordination in Mixed Unsignalized Intersections*. Revision under IEEE Transactions on Intelligent Transportation Systems, 2025.
- Felix Lokin and Fenghui Yu. *Fill Probabilities in a Limit Order Book with State-Dependent Stochastic Order Flows*. Submitted, arxiv:2403.02572, 2024.
- Feng-Hui Yu, Wai-Ki Ching, Chu-Fang Wu and Jia-Wen Gu. *Optimal Pairs Trading Strategies: a Stochastic Mean-Variance Approach*. Journal of Optimization Theory and Applications, 196, 36–55, 2023.
- Dong-Mei Zhu, Jia-Wen Gu, Feng-Hui Yu, Tak-Kuen Siu and Wai-Ki Ching. *Optimal Pairs Trading with Dynamic Mean-Variance Objective*. Mathematical Methods of Operations Research, 94(1), 145–168, 2021.
- Dong-Mei Zhu, Jia-Wen Gu, Feng-Hui Yu, Wai-Ki Ching and Tak-Kuen Siu. *How Correlation Risk in Basket Credit Derivatives Might be Priced and Managed?* IMA Journal of Management Mathematics, 32(2), 195–219, 2021.
- Feng-Hui Yu, Jie-Jun Lu, Jia-Wen Gu and Wai-Ki Ching. *Modeling Credit Risk with Hidden Markov Default Intensity*. Computational Economics, 54(3), 1213–1229, 2019.
- Feng-Hui Yu, Wai-Ki Ching, Jia-Wen Gu and Tak-Kuen Siu. *Interacting Default Intensity with Hidden Markov Process*. Quantitative Finance, 7(5), 781–794, 2017.
- Tak-Kuen Siu, Feng-Hui Yu, Qing-Qing Yang, Jia-Wen Gu and Wai-Ki Ching. *Pricing Participating Policies Under Hidden Markov Models via Neural Networks*. Preprint, 2019.

Co-organized Events

2026–Present	<i>Delft Institute of Applied Mathematics Lunch Colloquium - TU Delft, Netherlands</i>
June 2026	Minisymposium: <i>Advances in FinTech and Financial Decision-Making (I & II)</i> 13th Bachelier World Congress - Bologna, Italy
Aug 2025	Minisymposium: <i>Theory and Algorithms in Trading and Market Simulation</i> NUS Quantitative Finance Conference - Singapore
July 2024	Minisymposium: <i>Control, Optimization and Market Microstructure</i> 12th Bachelier World Congress - Rio de Janeiro, Brazil
Sept 2023	15th European Summer School in Financial Mathematics - Delft, Netherlands

Academic Talks

Sept 2026	6th International Conference on Computational Finance - Oxford, UK
June 2026	13th Bachelier World Congress - Bologna, Italy
Aug 2025	Summer School of the Dutch Sequential Decision-Making Community - Eindhoven, Netherlands
Aug 2025	NUS Quantitative Finance Conference - Singapore
July 2025	SIAM Conference on Financial Mathematics and Engineering - Miami, USA
June 2025	12th General AMaMeF Conference - Verona, Italy
May 2025	4TU.AMI.SRI Real-World-Inspired Sequential Decision Making Meeting - Utrecht, Netherlands
Nov 2024	BIRS Workshop on Modeling, Learning and Understanding: Modern Challenges between Financial Mathematics, Financial Technology and Financial Economics - Banff, Canada
July 2024	12th Bachelier World Congress - Rio de Janeiro, Brazil
June 2024	Statistics and Machine Learning in Finance Seminar - University of Oxford, UK
April 2024	Recent Advances in Stochastic Control, Machine Learning and Quantitative Finance - Shanghai, China
April 2024	5th International Conference on Computational Finance - Amsterdam, Netherlands
June 2023	4TU.Applied Mathematics Institutes Summer Event - Delft, Netherlands
Mar 2023	Dutch Mathematical Finance Seminars - TU Delft, Netherlands
Feb 2023	Seminar in Mathematical Finance - McMaster University, Canada
Nov 2022	14th DIAM Research Day - TU Delft, Netherlands
Oct 2022	Finance Research Day - TU Delft, Netherlands
Oct 2020	Seminar of Josef Teichmann's Group - ETH Zürich, Switzerland
Dec 2019	Youth Probability and Statistics Forum - Beijing, China
Dec 2019	Post/Doctoral Seminar in Mathematical Finance - ETH Zürich, Switzerland
Feb 2018	Workshop on Computational Science and Optimization - Hong Kong
Nov 2016	SIAM Conference on Financial Mathematics and Engineering - Austin, USA
Jul 2016	6th International IMS-FIPS Workshop - Edmonton, Canada
Jun 2016	11th East Asia SIAM Conference - Macao
Feb 2016	4th Asian Quantitative Finance Conference - Osaka, Japan

Teaching

(All of the courses were given in English)

Spring 2026	Lecturer , Delft Institute of Applied Mathematics - TU Delft, Netherlands <i>Financial Mathematics</i>
Spring 2026	Lecturer , Delft Institute of Applied Mathematics - TU Delft, Netherlands <i>Quantitative Risk Management</i>
Spring 2025	
Spring 2024	
Spring 2023	
Fall 2025	Lecturer , Delft Institute of Applied Mathematics - TU Delft, Netherlands <i>Calculus (for Computer Science Department)</i>
Summer 2025	Lecturer , Summer School of the Dutch Sequential Decision-Making Community (for the Landelijk Netwerk Mathematische Besliskunde (LNMB))- Eurandom, Netherlands <i>Foundations of Stochastic Optimal Control and Connections to Reinforcement Learning and Applications in Finance – From Stochastic Control to Algorithmic Trading</i>
Spring 2025	Lecturer , Delft Institute of Applied Mathematics - TU Delft, Netherlands <i>Special Topics in Financial Engineering (Machine Learning in Finance)</i>
Spring 2024	
Spring 2023	
Spring 2022	Teaching Assistant , RiskLab - ETH Zürich, Switzerland <i>Quantitative Risk Management</i>
Spring 2021	
Fall 2017	Teaching Assistant , Department of Mathematics - The University of Hong Kong, Hong Kong <i>Discrete Mathematics</i>

Fall 2017	Teaching Assistant , Department of Mathematics - The University of Hong Kong,
Fall 2016	Hong Kong
Fall 2015	<i>University Mathematics I</i>
Spring 2015	Teaching Assistant , Department of Mathematics - The University of Hong Kong,
	Hong Kong
	<i>Computational Methods and Differential Equations with Applications</i>

Supervision

PhD Student	Christian Tezza (Università di Bologna, jointly with MN for a period of 6 months), 2024
Master Students	<i>Delft Institute of Applied Mathematics, TU Delft, Netherlands</i> Severijn Wiechers (jointly with SwissLife), ongoing Konstantinos Giannakaras, ongoing Delvin Karjadi (jointly with MN), ongoing Mustafa Tugtekin (jointly with MN), ongoing Joep Claes, Aug, 2025 David Brouwers (jointly with MN), Oct, 2024 Giorgia Bifronte, Aug, 2024 Luca Fornaro (jointly with MN), Aug, 2024 Tomás Carrondo (jointly with Christa Cuchiero), June, 2024 Laurens Chin-A-Pauw, Mar, 2024 Felix Lokin (jointly with MN), Sept, 2023 Vangelis Nakos, Sept, 2023 <i>Department of Computer Science, TU Delft, Netherlands</i> Reinier Schep (jointly with Frans Oliehoek), ongoing Jorn Teurlings (jointly with Frans Oliehoek and Mustafa Mert Celikok), Aug, 2025
Bachelor Students	<i>Department of Computer Science, TU Delft, Netherlands</i> Mihail Bankov, Radoslav Georgiev, Tsvetelina Ilieva, Tudor Pagu, Cristian Petre-Luca (jointly with Frans Oliehoek), June, 2026

Additional Information

Mar 2025	Certificate of University Teaching Qualification - <i>4TU.Education, Netherlands</i>
2024–Present	Member of the EEMCS Faculty Graduate School PhD Mentor Group - <i>TU Delft, Netherlands</i>
Jan 2015	Certificate of Teaching and Learning in Higher Education - <i>The University of Hong Kong, Hong Kong</i>
Membership	Bachelier Finance Society, Society for Industrial and Applied Mathematics
Languages	Chinese (native), English (fluent)